MD. SAI MON HASAN EMON



Software Engineer | Full-Stack Developer & Backend Systems

Uttara, Dhaka, Bangladesh | saimonemon46@gmail.com | +880 1622 015799 | LinkedIn | GitHub | Portfolio

Software Engineer with experience in software development and building scalable web applications and backend systems using Python/FastAPI, Laravel, React/Next.js, and relational databases. Skilled in SQL performance, debugging, and containerized development with a focus on maintainable, reliable features.

AREAS OF EXPERTISE

Languages & Frameworks: Python, JavaScript, PHP, C++, Laravel, React.js, Next.js, Django, FastAPI, Tailwind CSS
Backend & AI/Data: REST APIs, JWT, PostgreSQL, MySQL, Redis, Docker, LangChain, LangGraph, RAG, FAISS, Pandas,
Tools & Practices: Git, GitHub, Linux, SQL Optimization, System Design, Problem Solving, Unit Testing

PROFESSIONAL EXPERIENCE

Software Engineer | IMJOL, Dhaka, Bangladesh | *Aug 2026 – Present*

- Develop client-facing features using Laravel and Next.js, translating business needs into production-ready code.
- Investigate bugs, analyze complex corner cases, and optimize PostgreSQL and Redis database operations.
- Utilize Docker, Linux environments, and Git workflows while collaborating with senior engineers on an agile team environment.

Software Engineer Intern | Leotech, Dhaka, Bangladesh | *Jan 2026 – May 2026*

- Contributed to an education management platform using HTML, CSS, Bootstrap, JavaScript, PHP, and MySQL.
- Implemented financial workflows in Academic and Fee modules covering admissions, collections, and discounts.
- Participated in code reviews and active debugging to maintain robust engineering standards.

SELECTED PROJECTS

LedgerCore – Payment Ledger Engine | GitHub
Tech Stack: Python, FastAPI, PostgreSQL, Redis, Docker

- Designed a double-entry payment ledger with concurrency control, deadlock prevention, and idempotent transaction handling; implemented automated concurrency and load tests to validate transaction correctness while adhering to SOLID principles.
- Built automated test coverage spanning isolated unit tests, PostgreSQL integration tests, API behavior, and concurrency stress scenarios.
- Tested double-spend prevention and concurrent idempotency handling, alongside Locust-based load testing.

Distributed Media Processing Pipeline | GitHub
Tech Stack: FastAPI, RabbitMQ, PostgreSQL, MinIO, Docker

- Architected asynchronous media-processing pipeline using background workers, message queues, and MinIO object storage.

UniMind – AI University Management System | GitHub
Tech Stack: React, Django, FastAPI, PostgreSQL, LangGraph, Docker

- Built multi-role management platform (students, faculty, advisors) with RBAC and AI-based performance analysis.

MediSA – AI Healthcare Triage & Recommendation | GitHub
Tech Stack: PHP, MySQL, FastAPI, LangChain, RAG, FAISS, Bootstrap

- Engineered AI-assisted platform for symptom triage and doctor matching. Named finalist in MillionX AI Buildathon (400+ teams).

PROBLEM SOLVING & EDUCATION

- Solved 100+ algorithmic problems across GeeksforGeeks (75+) and LeetCode (25+), covering Data Structure & Algorithms, dynamic programming, dynamic graph traversals, and search algorithms.

B.Sc. in Computer Science & Engineering – IUBAT | *CGPA: 3.84/4.00*

CERTIFICATIONS

• Applied Data Science & Machine Learning (Amar Ischool) | • Building NLP Projects (Amar ISchool)
• Prompt Engineering (Grameenphone Academy) | • MillionX AI Buildathon Finalist (MillionX)